Toward a Phenomenon-Preserving Calculus of Physical Representations

The Yang–Mills Mass Gap as a Problem of Representational Transport

Abstract

The fragmented perspectives on the Yang–Mills mass-gap problem point toward a need deeper than another choice of field variables. What is missing is a mathematics of the relations between representations: a framework that states which physical phenomena, structural conditions, and quantitative bounds are preserved when variables, regulators, scales, or apparent ontologies change. This essay develops a phenomenon-first conception of physical representation. It separates defining physical content, admissibility conditions, emergent targets, and auxiliary constructs; introduces representation packages and proof-bearing morphisms; and treats the mass gap as infrared spectral rigidity that must be generated in one representation and transported to the physical continuum spectrum. The resulting perspective reframes the search for a mass gap as the search for a representation in which non-Abelian dynamics becomes a uniform positive inequality together with certified transformations carrying that inequality to the reconstructed quantum theory.

1 The new representational need

The fragmented descriptions of Yang–Mills theory suggest a central conclusion:

> We do not merely need a better representation of Yang–Mills fields. We need a mathematical representation of the relations between representations—one that records which physical phenomena survive when variables, scales, regulators, and even apparent ontologies change.

The classical action

$$S[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \mathrm{Tr}(F_{\mu\nu} F^{\mu\nu})\, d^4x, \qquad F = dA + A \wedge A, \tag{1}$$

is an exceptionally economical representation of local geometric dynamics. It is not, however, a representation in which the expected infrared structure is manifest. The gauge potential is redundant, the classical action is scale-free, and perturbative excitations are massless, whereas the expected physical theory has a dynamically generated scale and a gauge-invariant massive spectrum.

Representing a theory by its preferred variables and equations therefore risks identifying a descriptive instrument with physical ontology. The mass-gap problem instead encourages the shift

$$\text{representation of objects} \quad \longrightarrow \quad \text{representation of physical invariants and their transformations.} \tag{2}$$

2 Why the usual notion of representation is insufficient

A conventional formulation normally selects fundamental variables, equations of motion, a state space, observables, and a quantization procedure. Beginning with the connection A_μ already carries several tacit commitments:

- the gauge potential is treated as the natural primitive object;
- locality is expected to be manifest at the level of primitive variables;
- the classical configuration space is presumed to remain the right ontology after quantization;
- the quantum theory is approached as a deformation of the classical theory;
- the particles visible around a perturbative vacuum are treated as suitable coordinates for the infrared regime.

These commitments need not all be false. The conceptual problem is that they may be poorly aligned with the target phenomenon. A representation can be faithful and calculationally powerful in one regime while being structurally blind to another. In Yang–Mills theory the classical and perturbative representations make the input law transparent while rendering the emergent infrared conclusion obscure.

3 A phenomenon ledger

The analysis should begin by separating four epistemic classes: defining physical content, structural admissibility conditions, target phenomena, and associated but logically distinct phenomena.

3.1 Defining physical content

The following features identify the intended theory or its physical universality class:

- four-dimensional continuum spacetime in the final theory;
- a compact simple gauge group G;
- the non-Abelian short-distance interaction structure;
- asymptotic freedom and its ultraviolet scaling;
- a sufficiently rich algebra of gauge-invariant observables and their correlation functions;
- nontrivial interaction.

Discarding these would risk solving a different problem.

3.2 Structural admissibility conditions

These conditions determine whether the constructed object is an acceptable quantum field theory:

- positivity of physical probabilities;
- unitarity after reconstruction;

- Poincaré covariance in the continuum;
- causal locality for physical observables;
- a stable vacuum and positive energy;
- compatibility of Euclidean and Hamiltonian descriptions;
- controlled continuum and infinite-volume limits.

They need not be manifest in every intermediate representation, but every temporary loss requires a recovery theorem.

3.3 Target phenomena

The following are conclusions to be generated rather than assumptions to be preserved:

- dimensional transmutation;
- a finite physical correlation length;
- exponential clustering;
- a positive spectral gap;
- a spectrum of massive gauge-invariant excitations.

This distinction guards against circularity. A construction that assumes a uniform exponential-decay estimate has already incorporated much of the desired mass-gap conclusion.

3.4 Associated but separate phenomena

Confinement, a Wilson-loop area law, flux-tube formation, detailed glueball masses, and topological susceptibility may help identify the dynamical mechanism. They are not interchangeable with the exact mass-gap statement. The logical implications among them require separate theorems.

4 Which constructs may be compromised?

A phenomenon-first approach distinguishes physical indispensability from historical familiarity.

The central distinction is between *manifest* and *recovered* structure. Locality may fail to be manifest in a dual variable while causal locality remains exactly reconstructible for physical observables. A lattice regulator may break continuous spacetime symmetry while the limiting theory recovers Poincaré covariance. The gauge potential can lose its status as a primitive without abandoning the gauge-invariant content of Yang–Mills theory.

5 Physical reality as an invariant intersection

The physical theory should not be identified with any one familiar representation. It may instead be conceived as the structure invariant across an atlas:

$$\begin{aligned}\mathfrak{R}_{\text{lattice}} &\longleftrightarrow \mathfrak{R}_{\text{Euclidean}} \longleftrightarrow \mathfrak{R}_{\text{observable}} \\ &\longleftrightarrow \mathfrak{R}_{\text{Hamiltonian}} \longleftrightarrow \mathfrak{R}_{\text{RG}} \longleftrightarrow \mathfrak{R}_{\text{dual}}.\end{aligned} \tag{3}$$

Construct	**Representational status**	**Physical content to preserve**
Gauge potential A_μ	Potentially replaceable	Gauge-invariant observable relations
Particular gauge fixing	Auxiliary	Independence of physical predictions from the gauge
Ghost fields	Representation-dependent	Correct physical state space and unitarity
Perturbative Fock vacuum	Potentially inadequate in the infrared	Existence and stability of the physical vacuum
Gluons as asymptotic particles	Not required in pure Yang–Mills theory	Gauge-invariant physical spectrum
Classical configurations as fundamental ontology	Potentially replaceable	Quantum observables and correlation structure
Manifest locality of primitive variables	Potentially relaxable	Causal locality of physical observables
Smooth continuum at intermediate stages	Regulatable	Exact recovery of the continuum limit
Lorentz symmetry at finite cutoff	Temporarily relaxable	Poincaré covariance in the limit
Path integral over A	One construction strategy	A positive consistent state on physical observables
Wilson loops as privileged primitives	Useful but not sacrosanct	The gauge-invariant information they encode
Specific renormalization scheme	Auxiliary	Universal scaling and scheme-independent physics
Particle ontology	Emergent	Spectral and scattering content where defined

Table 1: A construct-compromise analysis for Yang–Mills representations.

The analogy with coordinate charts on a manifold is helpful but incomplete. Transformations between physical representations can be singular, exact only in a limit, valid only within a scale regime, nonlocal, or irreversible under coarse-graining. They may change the apparent ontology and may preserve some observables while discarding others. Representations alone are therefore not enough: the transformations between them must become proof-bearing mathematical objects.

6 A phenomenon-preserving representation package

A representation may be organized as a typed package

$$\mathfrak{R} = (\mathcal{K}, \mathcal{A}_{\text{phys}}, \mathcal{S}, \mathcal{D}, \mathcal{G}, \mathcal{L}, \mathcal{Q}), \tag{4}$$

where:

$\mathcal{K}$ is the carrier or kinematic presentation: connections, lattice links, loops, operators, or dual variables;

$\mathcal{A}_{\text{phys}}$ is the algebra or designated class of physical observables;

$\mathcal{S}$	is the admissible state space;
$\mathcal{D}$	specifies dynamics, correlations, or time evolution;
$\mathcal{G}$	records physical symmetries, redundancies, equivalence relations, and constraints;
$\mathcal{L}$	specifies scale transformations and limiting operations;
$\mathcal{Q}$	contains quantitative control data: norms, errors, decay rates, and domains of uniformity.

The separation of $\mathcal{K}$ from $\mathcal{A}_{\text{phys}}$ is decisive. Two representations can have radically different carriers while encoding the same observable theory. Conversely, a formal change of variables that resembles an algebraic equivalence need not be physically adequate if it fails to preserve states, positivity, or the limiting structure.

7 Proof-bearing morphisms

Let

$$\Phi : \mathfrak{R} \longrightarrow \mathfrak{R}' \tag{5}$$

be a transformation between representations. It should include an observable pullback

$$\Phi^* : \mathcal{A}'_{\text{phys}} \longrightarrow \mathcal{A}_{\text{phys}} \tag{6}$$

and an appropriate transformation of states. Exact physical equivalence would require

$$\omega'(O') = \omega\big(\Phi^* O'\big) \tag{7}$$

for the relevant states and observables.

For a regulated, approximate, or scale-dependent translation, the morphism must instead supply a quantitative certificate

$$\big|\omega'(O') - \omega\big(\Phi^* O'\big)\big| \leq \varepsilon_\Phi(O'; a, L, \mu). \tag{8}$$

The decisive question is whether

$$\sup_{O' \in \mathcal{C}} \varepsilon_\Phi(O'; a, L, \mu) \longrightarrow 0 \tag{9}$$

for a convergence-determining class $\mathcal{C}$, with the uniformity required for the continuum and infinite-volume limits.

Every morphism should state its epistemic type: exact equivalence, gauge quotient, regularization, coarse-graining, reconstruction, continuum limit, asymptotic equivalence, duality, or heuristic correspondence. This prevents a numerical approximation, an effective theory, a conjectural duality, and an exact equivalence from being treated as though they carried the same force.

8 Phenomenon-preservation certificates

Transformations should carry explicit proof obligations.

Observable preservation. Does the morphism preserve expectation values and correlation functions for a sufficiently rich gauge-invariant observable class?

Positivity preservation. Does a positive Euclidean functional remain capable of reconstructing a positive physical Hilbert space?

Symmetry preservation. Which symmetries are exact, broken by the regulator, anomalous, or recovered only in the limit?

Locality preservation. Is locality manifest, weakened, or reconstructed? Can nonlocal primitive variables still yield causal locality of physical observables?

Scale preservation. Does the transformation retain the correct beta function, generated scale, and universality class?

Limit preservation. Are the bounds uniform in lattice spacing, volume, and observation distance? Can the relevant limiting operations be exchanged?

Spectral preservation. Does a decay, mixing, coercivity, or contraction estimate in one representation imply a positive spectral bound after physical reconstruction?

The last obligation is the core of the mass-gap problem.

9 The gap across representations

The same expected phenomenon has distinct mathematical manifestations.

Representation	**Candidate gap witness**
Hamiltonian	$H \geq \Delta(1 - P_0)$
Euclidean correlations	Uniform exponential clustering
Transfer operator	Separation between leading and subleading spectral values
Probability measure	Uniform mixing, Poincaré inequality, or log-Sobolev inequality
Renormalization group	Controlled flow into a massive regime with finite physical correlation length
Effective action	Quantitative coercivity around the physical vacuum
Observable algebra	Absence of energy-momentum spectral support near zero
Lattice theory	A gap estimate uniform in volume and positive in physical units as $a \to 0$

Table 2: Representation-dependent witnesses of infrared rigidity.

These witnesses are not automatically equivalent. The new framework must contain transfer theorems of the form

$$\text{certificate in } \mathfrak{R}_i \quad \Longrightarrow \quad \text{mass gap in } \mathfrak{R}_{\text{physical}}. \tag{10}$$

One quantitative form would be

$$\Delta_{\text{physical}} \geq c_i g_i - \varepsilon_i, \qquad c_i > 0, \tag{11}$$

where g_i is a representation-dependent rigidity witness and ε_i vanishes uniformly through the required limits.

10 From emergence to coercivity

In Hamiltonian language, the mass gap is a universal exclusion statement: no unknown sequence of physical states may have positive energies tending to zero. A more favorable representation could turn this negative statement into a positive inequality. Candidate forms include:

- a Poincaré or logarithmic Sobolev inequality;
- contraction of a transfer or coarse-graining operator;
- uniform probabilistic mixing;
- coercivity of a gauge-invariant effective action;
- a monotone renormalization-group quantity;
- a direct operator inequality above the vacuum sector.

For example, one might seek

$$\mathrm{Var}_\mu(f) \le \frac{1}{\lambda}\mathcal{E}_\mu(f,f), \qquad \lambda > 0, \tag{12}$$

and a theorem transporting λ to the physical spectral gap. Such a reformulation would turn collective emergence into mathematical rigidity. Its value depends entirely on proving that the inequality is gauge invariant, nonperturbatively valid, uniform in volume, stable under renormalization, correctly scaled in the continuum limit, and sufficient for spectral reconstruction.

11 A hierarchy of compromise and preservation

The framework imposes the following discipline:

May change freely

variables, gauges, regulators, coordinates, auxiliary fields

↓

May change with a recovery theorem

manifest locality, Lorentz symmetry, continuum form, particle language

↓

Must be preserved

physical observables, positivity, causality, universality class

↓

Must be derived

dynamical scale, exponential clustering, positive mass gap

(13)

This hierarchy guards against opposite errors. Preserving every familiar construct may trap the proof in a representation that hides the phenomenon. Relaxing defining physical structure may produce a gap for a different theory.

12 The deeper conceptual lesson

The physical content of Yang–Mills theory may reside neither in fields, particles, loops, nor lattice variables considered in isolation. It may reside in a network of stable relations across representations. On this view,

- gauge fields are coordinates on physical information;
- regulators are temporary realizations;
- particles are spectral interpretations;
- renormalization is a transformation of descriptive scale;
- the continuum theory is a coherence condition across limits;
- the mass gap is an invariant form of infrared rigidity.

The missing theory is thus a form of *representational dynamics*: a mathematics of how physical descriptions transform, what they forget, what they preserve, and which emergent properties become coercive in each one.

13 Central proposition

The research programme can be organized around the following proposition:

> The Yang–Mills mass-gap problem is fundamentally the problem of generating and transporting infrared spectral rigidity across a chain of representations whose natural variables separately expose locality, gauge invariance, positivity, scale dependence, and spectral structure.

This changes the search question from

> Which existing formalism can directly prove the mass gap?

to

> In which representation does non-Abelian quantum dynamics become a uniform positive inequality, and through which certified transformations can that inequality be transported to the physical continuum spectrum?

The missing object may therefore be neither a new field variable nor a new computational technique alone. It may be a phenomenon-preserving calculus of representations capable of turning collective quantum emergence into transferable mathematical rigidity.

References

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